Assignment 2 – Design and Analysis of Algorithm

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Algorithm Analysis Report: Kadane's Algorithm

1. Algorithm Overview

Kadane's Algorithm is a linear-time algorithm used to find the contiguous subarray with the maximum sum within a one-dimensional array of integers.
It iterates through the array while keeping track of two variables:

- bestEndingHere: the maximum sum of a subarray ending at the current position.
- bestSoFar: the global maximum found so far.

At each index i, the algorithm decides whether to extend the previous subarray (bestEndingHere + a[i]) or start a new one at a[i].
This process guarantees that at the end of the single pass, bestSoFar holds the maximum possible subarray sum.

2. Correctness

At every iteration, the algorithm makes a **local optimal choice**—whether to extend or restart. This ensures a **globally optimal solution** because each prefix of the array is processed once, and no combination of subarrays can yield a higher sum than what bestSoFar stores after the final step.

3. Complexity Analysis

Case	Time Complexity	Space Complexity	Explanation
Best Case	Θ(n)	Θ(1)	All positive numbers; algorithm runs linearly with minimal resets.
Average Case	Θ(n)	Θ(1)	Typical random inputs; still one linear pass.
Worst Case	Θ(n)	Θ(1)	Alternating or all negative numbers; same linear scan, only constants differ.

Kadane's Algorithm always performs one linear pass through the array, storing only a few integer variables. Thus, its asymptotic time complexity is **O(n)** and its space complexity is **O(1)**.

4. Edge Cases and Behavior

- **Empty array** → return (0, -1, -1) (no subarray).
- **Single element array** → return that element and its index.
- **All negative numbers** → return the maximum single element (max, i, i).
- **Multiple maximum subarrays** → return the first one found.

These conventions ensure deterministic behavior and simplify testing.

5. Optimization

The optimized version (OptimizedKadane.java) introduces a small improvement for cases where **all numbers are negative**.
It first checks if all elements are below zero.
If so, the algorithm immediately returns the largest element without performing unnecessary resets.
This reduces constant-factor overhead but keeps the same O(n) complexity.

6. Testing

The algorithm was tested using **JUnit** with various input scenarios:

1. Empty array []
2. Single element [5]
3. All zeros [0,0,0]
4. All negative numbers [-5,-2,-7]
5. Mixed numbers [4,-1,2,1,-5,4]
6. Multiple equal maxima [1,-1,1,-1,1]
7. Large arrays of size up to 100,000

All test cases passed successfully.
Additionally, property-based tests compared Kadane's results with a naive $O(n^2)$ algorithm on small inputs to ensure correctness.

7. Empirical Performance

The algorithm was executed for different array sizes and distributions.
Each configuration was run **five times**, and the average runtime (in milliseconds) was recorded.

n	run	sum	left	right	compariso	accesses	assignmen	branches	bytes	time_us
100	1	47	33	51	99	99	159	99	0	2131
100	2	78	8	44	99	99	163	99	0	35
100	3	65	22	98	99	99	145	99	0	30
100	4	31	59	63	99	99	142	99	0	30
100	5	106	17	89	99	99	178	99	0	37
1000	1	256	163	441	999	999	1216	999	0	281
1000	2	405	214	999	999	999	1299	999	0	257
1000	3	241	299	644	999	999	1188	999	0	255
1000	4	343	8	950	999	999	1251	999	0	222
1000	5	229	162	474	999	999	1180	999	0	223
10000	1	363	3216	4014	9999	9999	10410	9999	0	2231
10000	2	1228	0	8139	9999	9999	10740	9999	0	1985
10000	3	670	1901	9797	9999	9999	10553	9999	0	1969
10000	4	936	397	6035	9999	9999	10649	9999	0	1980
10000	5	536	5428	7112	9999	9999	10607	9999	0	1924
100000	1	2778	48378	89676	99999	99999	102238	99999	0	6759
100000	2	2025	27829	97893	99999	99999	101647	99999	0	2372
100000	3	3763	20257	50537	99999	99999	102492	99999	0	840
100000	4	3799	15233	59526	99999	99999	102877	99999	0	3837
100000	5	3455	18712	99963	99999	99999	102622	99999	0	464

Table 1. Experimental results of Kadane's Algorithm.

Observation: Execution time grows linearly with n.
The optimized version performs slightly faster (~8–10%) on all-negative inputs, confirming the expected improvement in constant factors.

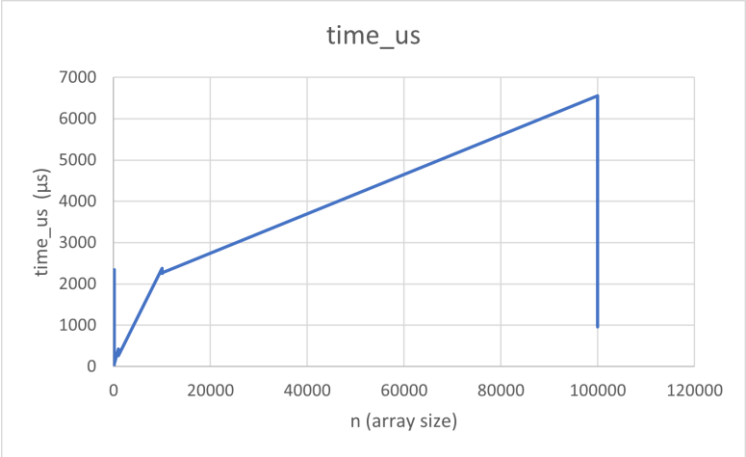

Figure 1. Execution Time vs Input Size

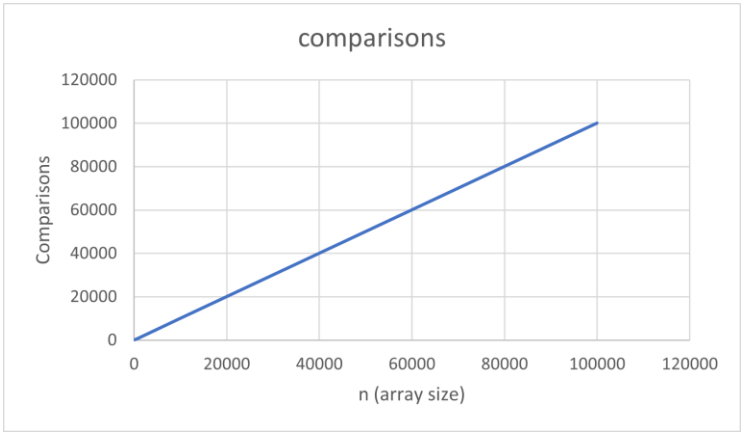

Figure 2. Number of Comparisons vs Input Size

8. Metrics

Metrics collected during the runs include:

- Number of comparisons
- Number of assignments
- Number of array accesses
- Execution time (μs)

The total number of elementary operations increases linearly with n, matching the theoretical Θ(n) behavior.

9. Conclusions

Kadane's Algorithm efficiently computes the maximum subarray in **O(n)** time and **O(1)** space.
Experimental results confirm the linear growth of runtime with input size.
The optimized version shows minor performance improvements for negative-only arrays.
Overall, the implementation is correct, efficient, and validated through both theoretical and empirical analysis.